UF/IFAS
Department of Wildlife Ecology and Conservation

3205 College Ave.
Davie, FL 33314-7719
352-659-1186

January 28th, 2026

Dr. Tara Cornelisse
Lead Editor
Natural Areas Journal

Dear Editor,

On behalf of my co-authors and myself, I would like to formally resubmit our manuscript titled "*Bioblitzes provides valuable biodiversity data*" for publication in the *Natural Areas Journal*.

We thank you for your consideration of the manuscript and the constructive reviews which helped us significantly improve this paper. We have listed all the referees' comments below and describe how we have responded to each suggestion. During the resubmission we uploaded a 'clean' version and a 'tracked changes' version to clearly highlight the changes made to our manuscript.

We hope that the revisions to the manuscript meet with your satisfaction, making this a valuable contribution to the *Natural Areas Journal*.

Sincerely,

Corey T. Callaghan (he/him)
Assistant Professor of Global Ecology
Department of Wildlife Ecology and Conservation
Fort Lauderdale Research and Education Center
University of Florida
w: www.coreytcallaghan.com
p: 352-659-1186
e: c.callaghan@ufl.edu

An Equal Opportunity Institution

UF UNIVERSITY of FLORIDA

Editorial Comments

The reviewers found your paper to be a valuable contribution to the literature, however there are some revisions requested. Reviewer #2 found the paper to be fairly well-written, while Reviewer #3 found it to be somewhat unorganized and repetitive. I recommend carefully reviewing the manuscript to remove instances of repetition and ensuring that any colloquial language is replaced with more scientific terminology. Please address the more specific comments of both reviewers as well, and make sure all figures are accessible, important, and well-explained. The comments from both reviewers are below; I look forward to receiving your revised manuscript.

Response: *We appreciate this thorough and positive set of reviews from the Reviewers. We have addressed each comment below, point by point, but highlight here the following overarching edits:*

- *We have revised the narrative so that it is clear it is (a) a framework and then (b) a case study using that framework. This included adding an additional paragraph at the end of the introduction focused on the framework itself.*
- *We edited the manuscript to remove repetition and minimize colloquial language throughout as suggested by Reviewer 2.*
- *We completed an additional exploratory analysis as suggested by Reviewer 1, comparing the results to more active samplin (e.g., at State Parks).*

UF UNIVERSITY of FLORIDA

Reviewer 2 Comments

This paper is a welcome addition to the literature. We have utilized bioblitzes in the past, and it is good to see that they are an excellent way at generating biodiversity data. I don't make this a requirement, but it would be interesting to compare the data you got to state parks or other protected lands that get regular visitors - so passive data collection vs active data collection. You can also look by taxon. These ideas may be beyond the scope of the paper, and I think the paper adequately demonstrates that bioblitzes do work.

Response: *We thank the Reviewer for these positive comments about our manuscript, and likewise hope it is a useful addition to the literature. Your idea is a good one! We actually had a recent paper which investigated the data from parks in Florida, and compared citizen science with state agency data (here: https://doi.org/10.1002/2688-8319.70173). We think this is a good idea by the Reviewer and as such we performed an exploratory analysis which used the parks from this paper (N=39) and put them in the context of the bioblitz (active sampling). We made an additional supplemental figure and added a couple lines about this to the discussion (Lines 284-289):*

> *"We further conducted an exploratory analysis, comparing DeLuca Bioblitz data to iNaturalist data from 39 Florida state parks (Lowe et al., 2026), to evaluate active versus passive sampling (Figure S5). Despite the DeLuca Bioblitz data being collected over just four days, compared to the full temporal extent of the state park data, the bioblitz yielded more observations and species than expected based on state park trends.".*

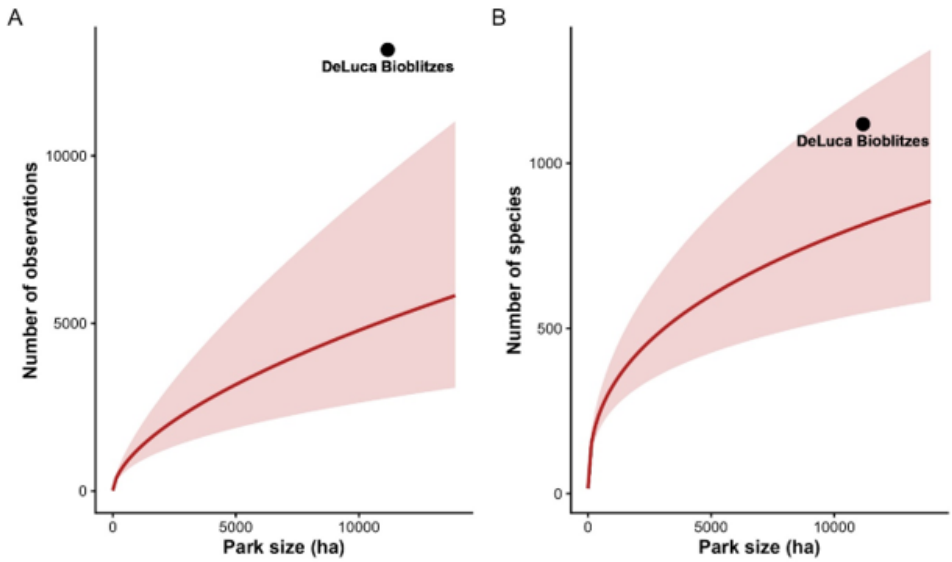

Figure S5. To explore how DeLuca Preserve Bioblitz observations compare to observations in Florida state parks, we modeled the relationship between number of observations and number of species across 39 state parks (see Lowe et al., 2026). We ran two negative binomial generalized linear models: one with the number of observations as the response and log-transformed park size (ha) as the predictor, and one with the number of species as the response and log-transformed park size (ha) as the predictor. Model structure was chosen based on residual diagnostics using the *DHARMa* package in R (Hartig, 2024). We then plotted the fitted trend lines with standard error, and highlighted data from the DeLuca Bioblitzes with a point showing where it falls on each graph. Note that the state park data covers all dates in iNaturalist, whereas DeLuca Preserve Bioblitz represents only observations taken during four bioblitzes. This plot illustrates the value of bioblitzes to collect more observations and species than passive observation.

I just have a few minor comments.

83. This is far too great of precision for the coordinates. 6 decimals is 43-111 mm, which is way too precise for a land area like this. Switching to 4 decimal places barely moved the marker. 4 is sufficient and 5 is the maximum.

Response: *We agree - we switched to 2 decimal places which is more than sufficient.*

163 - spell out numbers under 10 (here are two examples of 4). See also 181 "1 or 2 times". There are a few other examples as well.

Response: *Suggestion adopted.*

You have some accessibility issues with the figures. To check whether colors can be distinguished, use a color blindness simulator like Coblis: https://www.color-blindness.com/coblis-color-blindness-simulator/

Response: *Thank you! We have revised the figure color schemes accordingly.*

Fig. 1. You can't tell the difference between red and purple under monochromacy. Although rare as a disorder, many people print papers out in black and white. Consider using shape in addition to color.

Response: *Good suggestion – we have added shape as well to the figure.*

Fig. 5 Many of the colors blend together under different forms of color blindness. Use shape in addition to color.

Response: *Suggestion adopted.*

Figure S2 might not be accessible. The orange and blue are almost indistinguishable under monochromacy. If you keep the same colors, I suggest that you move the labels to the actual lines on the graph instead of in a legend below. You should also not refer to colors in the legend. Same will go for Fig. S4.

Response: *We have gone through and checked these all and edited accordingly. Thank you for this suggestion.*

Reviewer 3 Comments

Hoskins et al. make a case for the value of bioblitzes for science, conservation and community engagement. They describe their efforts in the DeLuca Preserve in Osceola County, Florida, over 4 years. This kind of project is great, further recognition of citizen science data and documentation of outcomes from bioblitzes like mentioned here is warranted.

Response: *We thank the reviewer for their positive assessment of the manuscript and for recognizing the value of bioblitzes for science, conservation, and community engagement. We agree that documenting the scientific outcomes of bioblitz efforts is important, and we hope this case study helps further highlight the broader utility of citizen science data for biodiversity research and management. We also are grateful for their comments below, which helped improved our manuscript.*

The ms. is not very well written or organized. The writing is highly repetitive, using up space where more interesting info could go. The writing style is a bit wordy, folksy in places and some passages stray from merely awkward into inaccuracy. With this many authors, someone should have taken the time to edit the submission.

Response: *We thank the reviewer for this feedback and have taken steps to improve clarity, and organization in several parts of the manuscript. Specifically, we have edited the Abstract, portions of the Introduction, and the opening sections of the Discussion to reduce redundancy, and improve the overall flow. We believe these targeted revisions improve readability and flow without altering the scope or conclusions of the study. We will highlight a select few edits here. We refer the Reviewer to see the revised manuscript uploaded in our revision for all changes.*

- *We trimmed the abstract, to remove the restatement of the value of a bioblitz.*
- *We removed the descriptive text about the user numbers which appeared in the methods and results. It now only appears in the results.*
- *We have added more interesting information (i.e., more background on the conceptual framework)*
- *We removed repetitive framing language in the Introduction that restated the general value of bioblitzes and citizen science across multiple paragraphs, tightening the narrative to focus more directly on the specific research gap addressed by this study.*
- *We also reworked the table describing/introducing the metrics to be stand alone, and separate from Figure 1.*
- *We removed redundant language, for example now only refer to regional scale as opposed to regional and landscape scale previously meaning the same thing.*
- *We revised the opening of the Discussion to explicitly enumerate the key findings of the study rather than reintroducing background material, reducing redundancy between the Introduction, Results, and Discussion.*
- *We edited parts of the Discussion to focus on flow, and allow some of the main conclusions to come at the end.*

The bioblitz outcomes are great and more should be done to highlight key findings. The "Overall contributions summary" section at top of results is a weak or off-track summary of their efforts. Slightly uncommon metrics pop up elsewhere in ms also.

Response: *We thank the reviewer for highlighting the strength of the bioblitz outcomes and for this constructive suggestion. In response, we revised the Results and Discussion to more clearly foreground the key findings of the study, including the detection of rare, endemic, and introduced species and the disproportionate contribution of bioblitz data to regional biodiversity benchmarks. We added the following to the Discussion first paragraph to better highlight the key findings (Lines 242-245):*

> *"Using a four-year case study, we show that a concentrated bioblitz effort can (1) generate dense, taxonomically diverse datasets over short time frames, (2) detect rare, endemic, and introduced species of management relevance, and (3) contribute disproportionately to regional biodiversity records.".*

We also revised the opening Results subsection to emphasize biodiversity outcomes rather than event-level descriptors and renamed the section "Overview of biodiversity outcomes". We removed the event description text, as this was already stated in the results. Ultimately, we feel this helps the narrative and keeps it more on track throughout.

We understand the comment about slightly uncommon metrics. We have now thought critically about the narrative and made some edits, which helps alleviate this concern. See the comment below for details.

I like the comparison to numerous random polygons in the state, not sure why the section is called "Gap Filling". The idea is not as novel as they suggest. Additional discussion of the obvious criticisms of this approach would strengthen their conclusions. E.g. do underlying biodiversity, observer access, quality or motivation play any role? Any drawbacks to comparing the average of random polygons?

Response: *We thank the reviewer for this helpful feedback. In response, we renamed the section formerly titled "Gap Filling" to "Comparison with random landscape benchmarks" to more accurately reflect the purpose of this analysis as a contextual comparison rather than a claim of novelty. We also revised the text to avoid overstating novelty and to clarify that the random polygon approach is intended as a benchmarking exercise to place bioblitz-derived biodiversity data into a broader landscape context. In addition, we expanded the Discussion to explicitly acknowledge limitations, which reads (Lines 279-282):*

> *"However, we acknowledge that underlying biodiversity, habitat composition, and accessibility vary across polygons, and observer behavior on iNaturalist is not spatially random, all of which can contribute to variation among polygons. As such, these comparisons are best interpreted as coarse contextual benchmarks rather than direct ecological contrasts.".*

I felt a bit of a mismatch between the narrative and figures. The biodiversity metrics in Fig 1. perhaps should be better explained or integrated into text. Some figure legends are not self-explanatory.

Response: *This comment made us think critically about the narrative. Our main fix here to focus the narrative was to allow the 'framework' and 'analysis' to be complementary and not at odds with one another. We first made this clear by editing the last paragraph of the introduction (Lines 65-70):*

> *"First, we provide a replicable framework that can be expanded outside of our study site, with the aim of helping natural resource managers leverage bioblitzes to advance biodiversity data collection. We then use this framework to illustrate the scientific value of a bioblitz as a mechanism for generating biodiversity data, including baseline species inventories, novel species records, and indicators of regional biodiversity utility (Figure 1).".*

We then added an additional paragraph to the end of the introduction, following the 'goals' of the paper, which solely introduces the framework (Lines 74-87):

> *"Bioblitz biodiversity utility framework*
> *We present a conceptual framework for evaluating the biodiversity value of bioblitzes by organizing metrics along two dimensions: their biodiversity utility for management and monitoring, and the analytical complexity required to calculate them (see Figure 1). The framework distinguishes site-level metrics that characterize biodiversity patterns within the site defined for the bioblitz (this includes, total observations, number of species, number of observers, and new species) from regional-level metrics that contextualize bioblitz observations relative to broader regional biodiversity data (this includes, relative frequency, relative rarity, and relative diversity of species, sampling intensity, and nearest external observation). Together, these metrics capture complementary dimensions of biodiversity value, ranging from baseline inventory completeness to the contribution of short-term, concentrated effort to regional biodiversity knowledge. Because we separate site-level and regional-level metrics, this framework provides a flexible tool that natural resource managers can adapt to evaluate the biodiversity outcomes of bioblitzes across different regions, taxa, and management objectives.".*

This allows the reader to then be prepared for the biodiversity metrics discussed in more detail in the methods, and subsequently, the results. This explicitly addresses the previous comment, about uncommon metrics, by helping the flow be more natural, aligning the narrative and figures, especially Figure 1, which sets up the narrative for the rest of the paper. In addition, we split Figure 1 into a figure and a more comprehensive Table which introduces the metrics, utlimately improving the flow.

The key result of 1252 species (line 113) in 11,000 hectares is great, perhaps a bit low for a

preserve of that size, in that area, I suspect they will add many, many more. Line 174 mentions 892 (research grade) and it looks like it is up to ~1800 in Figure 3?

Response: *There was a mistake in Figure 3 which has now been fixed. We have now focused on the Research Grade total, as this makes the most conceptual sense to focus on. We also agree that more will be added, and revised the text accordingly (Lines 193-195):*

> *"with each sampling effort and that sampling completeness has not yet been achieved (Figure 3), suggesting that more species will be found with ongoing efforts.".*

Potential typos / minor suggestions:

line 122 - perhaps more than 200 million?

Response: *Yes – good catch! We have fixed this. Thank you.*

line 125 - any observations can be downloaded, not only research grade

Response: *Yes, true. We have edited this sentence to be more precise (Lines 132-135):*

> *"All observations are made available for download on iNaturalist and Research Grade observations are uploaded to the Global Biodiversity Information Facility (GBIF), depending on license requirements, where they are made available to scientists around the world.".*

line 157 - mention R as well

Response: *Good suggestion. We added the following section shortly after this placement to not break up the flow too much (Lines 176-177):*

> *"All data analysis was conducted in R (R Core Team 2025) and relied heavily on the Tidyverse workflow (Wickham et al. 2019).".*

line 169 - github link broken or still private

Response: *We have made it public! And will archive the repository on Zenodo once accepted.*